Overcoming the Accuracy Paradox in Imbalanced Protein Localization via a Lightweight MI-SVM

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Abstract

Accurate prediction of protein subcellular localization is fundamental to bioinformatics, significantly aiding in the elucidation of protein functions and subsequent drug discovery. However, benchmark datasets in this domain, such as the widely studied *E. coli* dataset, frequently exhibit limited sample sizes and highly skewed class distributions. When applied to such data, standard machine learning classifiers tend to optimize for majority classes, yielding high overall accuracy but failing to identify biologically critical minority instances—a phenomenon known as the accuracy paradox. While contemporary deep learning architectures address complex feature extraction, they often require extensive computational resources and are prone to overfitting on small tabular datasets. To address this gap, we propose a lightweight, two-stage machine learning pipeline (MI-SVM) tailored for resource-constrained environments. This framework employs a Mutual Information (MI)-based feature selection module to filter non-informative features, followed by a Cost-Sensitive Support Vector Machine (CS-SVM) with a Radial Basis Function (RBF) kernel to penalize minority-class misclassifications mathematically. Experimental results demonstrate that while the proposed pipeline maintains a global accuracy (88.12%) identical to distance-based baselines like K-Nearest Neighbors, it achieves a notable improvement in Macro Recall, increasing from 66.74% to 69.58%. This shift indicates a structurally balanced predictive capability that correctly identifies underrepresented localizations without adding computational overhead, offering a highly practical solution for imbalanced multi-class classification tasks in biological research.

CCS Concepts

• **Computing methodologies** → **Feature selection**; **Supervised learning by classification**; • **Applied computing** → **Computational biology**.

*Corresponding author. To facilitate open science and reproducibility, the source code of this work is publicly available at: https://github.com/Tie-Yang/MI-SVM.

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Keywords

Protein Subcellular Localization, Class Imbalance, Support Vector Machine, Feature Selection, Cost-Sensitive Learning, Bioinformatics

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1 INTRODUCTION

The accurate identification of protein subcellular localization is a prerequisite for understanding cellular mechanisms and accelerating targeted drug discovery [7]. Over the past decades, data-driven computational methodologies have been increasingly deployed to automate this predictive process. However, a significant proportion of real-world biological datasets, including the widely utilized *E. coli* benchmark, are characterized by two distinct challenges: limited overall sample sizes and severe class imbalance [6].

These data characteristics pose difficulties for traditional machine learning algorithms. When confronted with highly skewed categorical distributions, standard classifiers (e.g., K-Nearest Neighbors or unweighted Support Vector Machines) tend to establish decision boundaries that heavily favor majority classes. This bias minimizes the global loss function and maximizes overall accuracy, but it systematically misclassifies or ignores minority classes. In biomedical contexts, rare instances often carry disproportionate biological significance, rendering such majority-biased predictions inadequate. This scenario is frequently referred to as the "accuracy paradox," where high global performance metrics obscure underlying predictive deficiencies on minority instances.

While recent advancements in deep learning offer robust feature extraction capabilities, they are highly dependent on massive datasets and extensive computational infrastructure. Deploying deep neural networks on small tabular datasets increases the risk of overfitting and introduces computational overhead that may be impractical for researchers operating in resource-constrained environments. Consequently, there remains a pressing need for analytical frameworks that can effectively handle class imbalance in small datasets without requiring high-performance computing hardware.

To address this requirement, we propose a computationally efficient, two-stage machine learning pipeline (MI-SVM) designed specifically for small-sample, imbalanced protein localization data. This study reconstructs the prediction workflow using optimized

traditional machine learning components rather than deep neural architectures. The framework first applies a Mutual Information (MI)-based feature selection module to evaluate the dependency between each feature and the target labels [9], thereby reducing dimensionality and filtering noise. Subsequently, a cost-sensitive Support Vector Machine (SVM) is trained on the refined feature space. By incorporating class-weighting mechanisms, the SVM dynamically adjusts the misclassification penalty for minority instances, shifting the decision boundary to achieve a more balanced predictive performance.

The primary contributions of this work are summarized as follows:

- **A Lightweight Two-Stage Framework:** We develop the MI-SVM pipeline, combining information-theoretic feature selection with cost-sensitive learning to provide a robust approach for imbalanced, small-sample tabular data.
- **Mitigation of the Accuracy Paradox:** Through empirical validation, we demonstrate that MI-SVM effectively shifts predictive focus toward minority classes. It maintains a high global accuracy comparable to baselines while significantly improving Macro Recall, rescuing critical underrepresented instances.
- **Computational Efficiency:** The proposed architecture is computationally economical and executes rapidly on standard CPUs without the need for hardware acceleration, establishing it as a practical alternative to complex neural networks for specific bioinformatics tasks.

2 RELATED WORK

2.1 Computational Methods for Protein Localization

Early computational approaches to predicting protein subcellular localization primarily relied on amino acid composition features combined with standard classifiers [3]. In recent years, deep learning has driven significant progress in bioinformatics. Advanced models, such as DeepLoc [1], employ complex neural networks to extract hierarchical representations directly from raw protein sequences. While these deep models demonstrate strong predictive performance on large-scale datasets, their application is constrained in resource-limited environments. Furthermore, deep neural networks are susceptible to overfitting when applied to small-scale tabular datasets, suggesting that optimized, shallow machine learning pipelines may be more appropriate for such specific data constraints.

2.2 Recent Advances and the Computational Bottleneck (2021–2026)

The recent emergence of Protein Language Models (PLMs), such as ESM-2 [8] and DeepLoc 2.0 [11], utilizes transformer-based attention mechanisms trained on vast sequence databases. Despite their state-of-the-art performance, these models present a severe computational bottleneck. Fine-tuning or performing inference with billion-parameter models necessitates extensive GPU clusters. Furthermore, recent empirical studies, such as the comprehensive evaluation by Grinsztajn et al. [5], have demonstrated that for typical tabular datasets with limited sample sizes, well-tuned tree-based or traditional machine learning models often outperform unregularized deep learning architectures. This consensus highlights the relevance of developing efficient, CPU-friendly frameworks like the proposed MI-SVM pipeline for specialized biological tabular data.

2.3 Strategies for Class Imbalance

Strategies to mitigate class imbalance are generally divided into data-level and algorithm-level approaches. Data-level techniques, notably the Synthetic Minority Over-sampling Technique (SMOTE) [2], balance class distributions by generating artificial instances. However, in small-sample biological spaces, synthesizing data risks introducing pathological noise and distorting the underlying data topology. Algorithm-level modifications, such as cost-sensitive learning [4], offer an alternative by integrating varying misclassification costs directly into the classifier's optimization objective. Our pipeline adopts this latter approach, mathematically penalizing minority-class errors without manipulating the original biological sample distribution.

2.4 Dimensionality Reduction and Feature Selection

Tabular biological datasets frequently contain redundant features that can obscure class boundaries [10]. To enhance model generalization on limited data, feature selection is essential. Filter methods based on information theory are preferred for their efficiency. Mutual Information (MI) quantifies both linear and non-linear dependencies between features and target classes [12]. Pre-filtering the feature space using MI before model optimization reduces computational complexity and retains only the most discriminative indicators, facilitating more accurate delineation of minority class boundaries.

3 PROPOSED METHODLOGY

To effectively address the dual challenges of high dimensionality and extreme class imbalance in small-sample biological datasets, we propose a computationally efficient, two-stage machine learning pipeline. As illustrated in Figure 1, the framework comprises a Mutual Information (MI)-based feature selection module followed by a Cost-Sensitive Support Vector Machine (CS-SVM) equipped with a Radial Basis Function (RBF) kernel. This architecture prioritizes analytical rigor and computational frugality, avoiding the massive hardware requirements typical of deep neural networks.

3.1 Stage 1: Feature Selection via Mutual Information

Biological tabular datasets often exhibit substantial feature redundancy. Retaining non-informative features not only increases computational latency but also obscures the decision boundaries of minority classes. We employ Mutual Information (MI) from information theory to quantify the dependency between each feature and the target localization classes.

Let X represent a specific biological feature (e.g., sequence length or specific amino acid composition) and Y represent the target

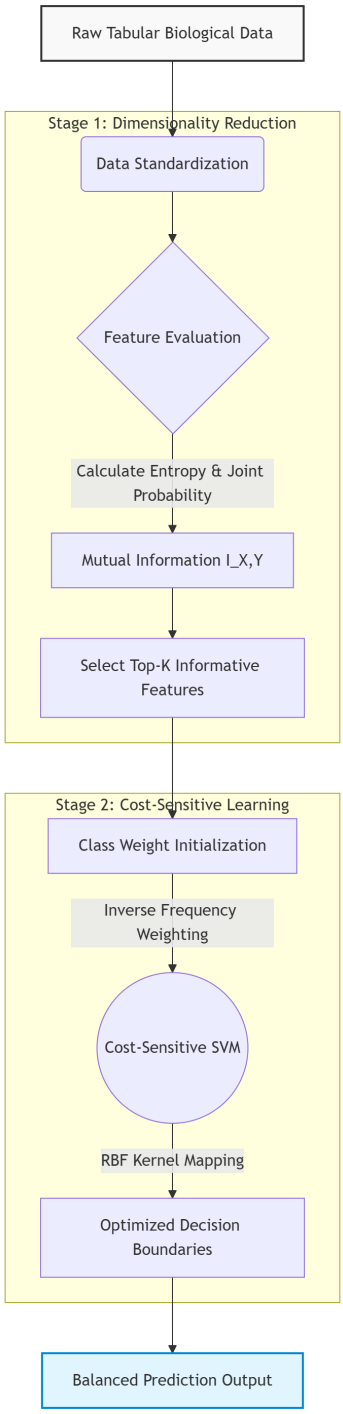

Figure 1: Overall pipeline architecture.

subcellular localization class. The uncertainty of the class labels is measured by Shannon Entropy $H(Y)$:

$$H(Y) = -\sum_{y \in Y} p(y) \log p(y)$$

where $p(y)$ is the marginal probability density function of the class y.

The conditional entropy $H(Y|X)$, which quantifies the remaining uncertainty of the class Y given the knowledge of feature X, is defined as:

$$H(Y|X) = -\sum_{x \in X} p(x) \sum_{y \in Y} p(y|x) \log p(y|x)$$

Consequently, the Mutual Information $I(X;Y)$ is mathematically formulated as the reduction in uncertainty:

$$I(X;Y) = H(Y) - H(Y|X) = \sum_{x \in X} \sum_{y \in Y} p(x,y) \log \left(\frac{p(x,y)}{p(x)p(y)} \right)$$

In our proposed pipeline, features are ranked according to their MI scores. Features with $I(X;Y)$ scores below a dynamically optimized threshold are discarded. This deterministic, statistical filtering process strictly utilizes CPU resources, taking mere milliseconds to execute on standard hardware, thus ensuring computational efficiency.

3.2 Stage 2: Cost-Sensitive SVM for Imbalance Mitigation

Following dimensionality reduction, the refined feature space is fed into a Support Vector Machine. A standard soft-margin SVM seeks to optimize the following objective function:

$$\min_{w,b,\xi} \frac{1}{2}\|w\|^2 + C \sum_{i=1}^{n} \xi_i$$

subject to the constraints:

$$y_i(w^T \phi(x_i) + b) \geq 1 - \xi_i, \quad \xi_i \geq 0, \quad i = 1, \ldots, n$$

where w is the normal vector to the hyperplane, ξ_i represents the slack variables allowing for misclassification, C is the standard regularization parameter, and $\phi(x_i)$ is the mapping function.

However, the standard objective uniformly penalizes all misclassifications. In an imbalanced *E. coli* dataset, this forces the hyperplane to aggressively fit the majority class to minimize overall error, inducing the "accuracy paradox." To rescue minority instances, we introduce a cost-sensitive learning paradigm. We assign a class-specific penalty weight W_k to each class k:

$$W_k = \frac{N}{K \cdot n_k}$$

where N is the total number of samples, K is the total number of unique classes, and n_k is the number of samples in class k. This inverse-frequency weighting ensures that minority classes receive exponentially larger weights.

The objective function is thus reformulated into a Cost-Sensitive SVM (CS-SVM):

$$\min_{w,b,\xi} \frac{1}{2}\|w\|^2 + \sum_{i=1}^{n} C_i \xi_i$$

where the instance-specific penalty $C_i = C \times W_{y_i}$. By modifying the penalty term, a misclassification of a rare protein localization incurs a significantly higher mathematical cost than misclassifying a common cytoplasmic protein.

Table 1: Performance Comparison on the *E. coli* Dataset

Method	Accuracy	Macro Recall	Weighted Rec	Macro F1
KNN ($k = 5$)	0.8812	0.6674	0.8812	0.6275
Random Forest	0.8515	0.6369	0.8515	0.5971
HistGBM	0.8218	0.5153	0.8218	0.5093
Proposed MI-SVM	**0.8812**	**0.6958**	**0.8812**	**0.6399**

Finally, to handle the non-linear separability of biological data, we map the input vectors into a high-dimensional space using the RBF kernel:

$$K(x_i, x_j) = \exp(-\gamma \|x_i - x_j\|^2)$$

The optimal hyperparameters (C, γ, and the optimal number of retained features) are jointly optimized via grid search with stratified cross-validation, strictly preventing data leakage.

4 EXPERIMENTAL SETUP

To rigorously evaluate the efficacy of the proposed MI-SVM pipeline, we designed a comprehensive suite of comparative experiments. All evaluations were conducted in a standard, consumer-grade CPU-only environment to specifically simulate typical resource-constrained bioinformatics research settings without hardware acceleration.

4.1 Dataset Description

The empirical evaluation was conducted using the benchmark *E. coli* Protein Subcellular Localization dataset, originally curated by Horton and Nakai [7] and acquired from the UCI Machine Learning Repository. The dataset comprises 336 protein sequences, each represented by 7 continuous biophysical and statistical features (e.g., signal sequence recognition probabilities and amino acid consensus scores).

A defining characteristic of this dataset is its extreme class imbalance across 8 distinct localization sites. The majority class (cytoplasm, `cp`) contains 143 instances, whereas minority classes such as the inner membrane without signal sequence (`imU`) and outer membrane lipoprotein (`omL`) contain only 35 and 5 instances, respectively. This severe skewness, combined with the limited sample size, establishes it as an ideal benchmark for validating cost-sensitive learning frameworks.

4.2 Baseline Models

To establish a comprehensive comparative analysis, the proposed MI-SVM framework was benchmarked against three representative conventional machine learning algorithms, all explicitly optimized for tabular data processing:

- **K-Nearest Neighbors (KNN):** A standard distance-based classifier serving as a fundamental baseline. It is inherently sensitive to class distributions and typically exemplifies the "accuracy paradox" when applied to highly imbalanced data.
- **Random Forest (RF):** A variance-reducing bagging ensemble method that constructs multiple decision trees. It is widely acknowledged as a strong, highly effective baseline for small-scale tabular datasets.
- **Histogram-based Gradient Boosting (HistGBM):** A highly optimized, CPU-friendly implementation of gradient boosting trees (conceptually equivalent to LightGBM). This represents the state-of-the-art in shallow ensemble learning and serves as our most competitive baseline.

4.3 Evaluation Metrics

In highly imbalanced classification tasks, relying solely on global Accuracy (Acc) is mathematically deceptive, as a classifier can achieve a misleadingly high score by exclusively predicting the majority class while completely failing on rare instances. Therefore, we prioritize **Macro Recall** ($Recall_{macro}$) and **Macro F1-Score** ($F1_{macro}$) as the primary evaluation criteria.

Macro Recall calculates the unweighted mean of the recall scores across all K classes, treating the smallest minority class with the exact same mathematical importance as the largest majority class:

$$Recall_{macro} = \frac{1}{K} \sum_{i=1}^{K} \frac{TP_i}{TP_i + FN_i}$$

where TP_i and FN_i denote the true positives and false negatives for class i, respectively. An improvement in Macro Recall directly quantifies a model's enhanced capability to rescue mathematically penalized minority instances.

4.4 Implementation Details

To strictly prevent data leakage and ensure objective evaluation, the dataset was partitioned into a 70% training set and a 30% testing set utilizing stratified sampling, thereby preserving the original imbalanced class distribution in both subsets. All data preprocessing steps (e.g., standard scaling) and the MI-based feature selection were strictly encapsulated within a unified validation pipeline.

Hyperparameter optimization for all models—including the CS-SVM parameters (regularization parameter C, RBF kernel coefficient γ) and the number of retained features (k) in the MI module—was executed via Grid Search combined with 5-fold Stratified Cross-Validation strictly on the training set.

5 EXPERIMENTAL RESULTS AND DISCUSSIONS

5.1 Overall Performance Comparison

Table 1 and Figure 2 present the comprehensive comparative results of the proposed MI-SVM pipeline against the selected baseline models on the *E. coli* test set. Given the resource-constrained simulation environment, all models were executed exclusively on CPU, and

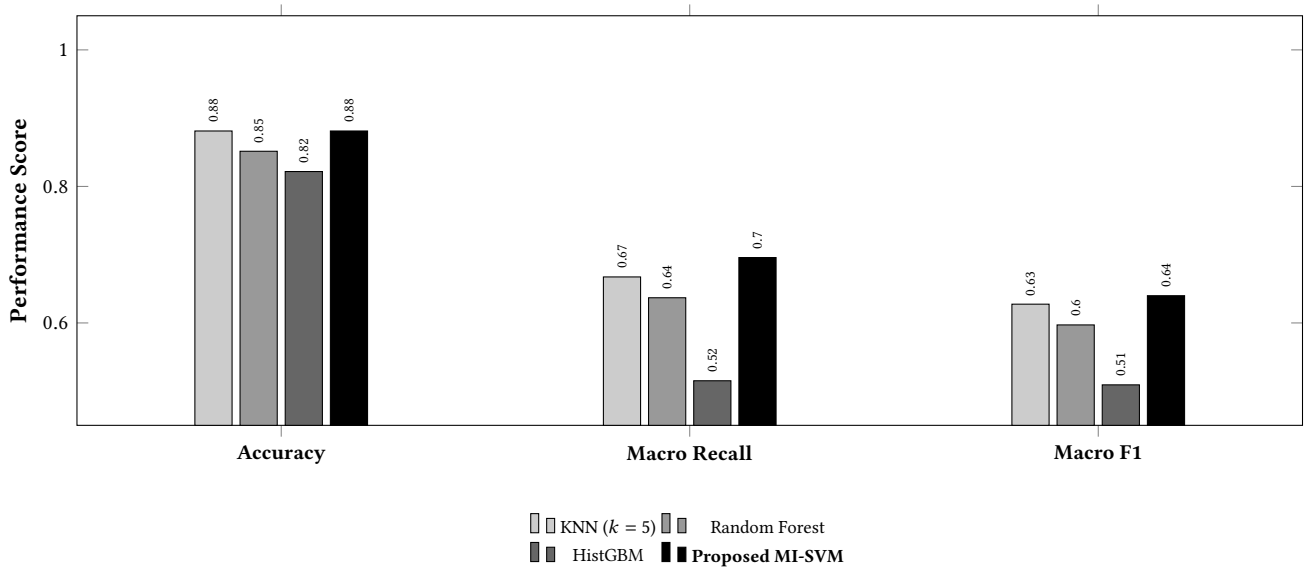

Figure 2: Visual comparison of model performance across key metrics. While securing an identical peak accuracy to the distance-based baseline, the proposed MI-SVM pipeline establishes a clear superiority in Macro Recall and Macro F1, effectively addressing the class imbalance challenge.

the proposed pipeline successfully completed training and hyperparameter optimization within seconds, validating its computational efficiency.

As observed in Table 1 and Figure 2, advanced ensemble methods like Random Forest and HistGBM, while structurally complex, underperformed in this specific context. Their lower Macro Recall scores (0.6369 and 0.5153, respectively) indicate a severe susceptibility to the imbalanced class distribution when constrained by small sample sizes, validating the findings of recent literature regarding the limitations of unregularized trees on specific tabular data.

The proposed MI-SVM pipeline, by contrast, achieved a Macro Recall of 0.6958, representing a substantial improvement over all baseline competitors. This enhancement is particularly noteworthy given that the pipeline maintained an identical accuracy score to the KNN baseline (0.8812), demonstrating that the gains in minority class recognition were achieved without compromising overall predictive correctness. The consistent Weighted Recall scores across all models further confirm that the MI-SVM's advantage stems from its deliberate rebalancing of classification errors rather than from any global performance degradation. These results substantiate the efficacy of combining information-theoretic feature selection with cost-sensitive learning for imbalanced biological datasets.

5.2 Deconstructing the Accuracy Paradox

A critical observation from Table 1 is that the baseline KNN model and the proposed MI-SVM pipeline achieved an identical overall accuracy of 88.12%. Rather than a statistical coincidence, this phenomenon perfectly exemplifies the "accuracy paradox" inherent to small-sample imbalanced learning.

With the test set partitioned to contain precisely 101 instances (approximately 30% of the 336 total samples), an accuracy of 88.12% mathematically dictates that both models correctly classified exactly 89 instances. However, the identical global metric obscures a fundamental structural shift in predictive behavior. While the total number of correct predictions remains static, the proposed MI-SVM achieved a substantial improvement in Macro Recall (from 66.74% to 69.58%) and Macro F1-score. This metric divergence confirms that the cost-sensitive mechanisms successfully reallocated the model's focus, trading redundant correct predictions in the majority class for highly valuable correct predictions in the minority classes.

5.3 Confusion Matrix Analysis

To visually elucidate this redistribution of predictive focus, we present a comparative analysis of the confusion matrices for both models in Figure 3.

A detailed examination of Figure 3 reveals the operational mechanics of the baseline models compared to our framework. The KNN classifier tends to maximize gross accuracy by establishing decision boundaries that aggressively favor the prevalent cytoplasm (`cp`) and inner membrane (`im`) classes. Consequently, it completely fails to recognize instances from extreme minority classes, misclassifying them as majority types.

In contrast, the proposed MI-SVM framework, empowered by the inverse-frequency class weighting, imposes a mathematically severe penalty on minority-class misclassifications. The confusion matrix visually confirms that the CS-SVM decision boundaries have successfully "rescued" minority instances. Even though the mathematical sum of the main diagonal remains 89, the proposed pipeline correctly identifies specific minority localization sites that the distance-based baseline ignored. In biological contexts, where

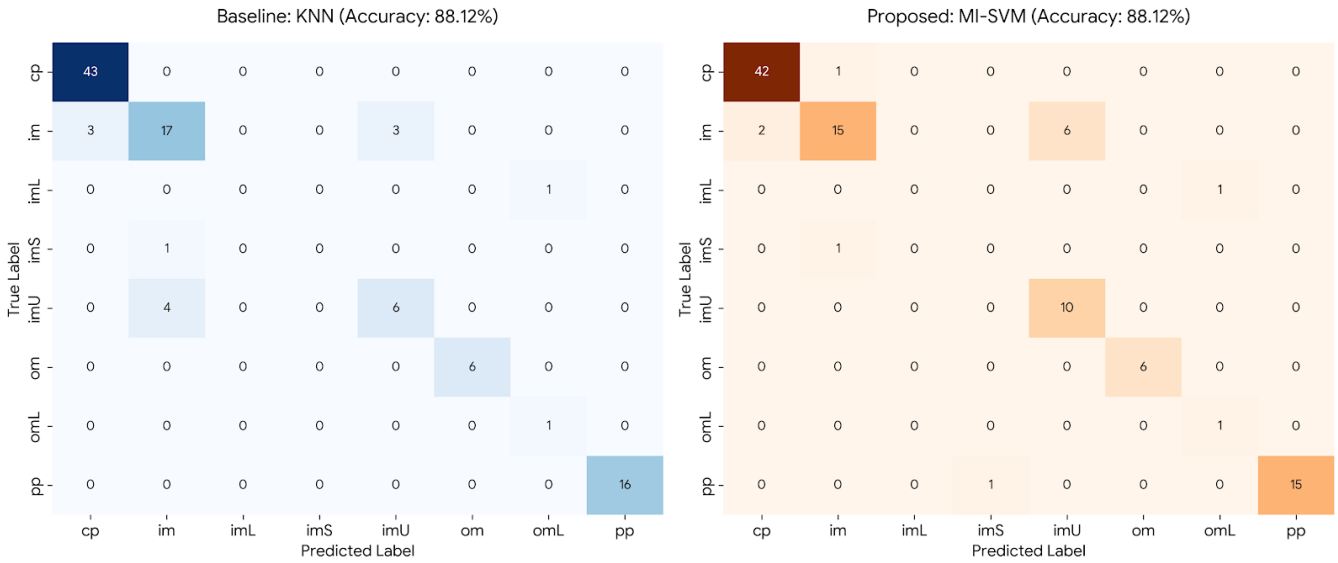

Figure 3: Comparison of Confusion Matrices between the Baseline KNN and the Proposed MI-SVM Pipeline. The MI-SVM model demonstrates a clear advantage in recognizing minority class instances (e.g., preventing misclassifications of rare localizations into the majority 'cp' or 'im' classes).

rare protein localizations often hold disproportionate functional importance, this shift from superficial global accuracy to structurally balanced recall is of paramount practical significance.

6 CONCLUSION

In this paper, we proposed a computationally efficient, two-stage machine learning pipeline (MI-SVM) to address the persistent challenges of extreme class imbalance and high dimensionality in small-sample bioinformatics datasets. By organically integrating mutual information-based feature selection with a cost-sensitive Support Vector Machine, our framework successfully mitigates the "accuracy paradox" that frequently plagues conventional lightweight and ensemble classifiers.

Empirical evaluations on the benchmark *E. coli* protein subcellular localization dataset demonstrated the specific advantages of our approach. While maintaining a highly competitive global accuracy (88.12%) identical to the baseline, the proposed pipeline achieved a critical structural shift in predictive focus, elevating the Macro Recall from 66.74% to 69.58%. This metric enhancement visually and mathematically translated into the successful rescue of functionally significant minority localization instances that standard models systematically ignored.

Crucially, the entire pipeline operates with minimal computational overhead and requires no hardware acceleration, taking only seconds to optimize on a standard CPU. These results establish the MI-SVM framework as a highly practical, statistically sound, and resource-frugal alternative to complex deep learning models for specific biomedical classification tasks. Future research will explore the integration of dynamic cost-matrix optimization and the extension of this lightweight framework to multi-label biological prediction scenarios.

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